

Maternal & Child Health Analysis

Patricia Mugabe

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1. Introduction

1.1 Background

Maternal and child health remains one of the most critical areas of public health concern in sub-Saharan Africa, where the burden of adverse pregnancy outcomes disproportionately affects low- and middle-income countries. Uganda, like many countries in the region, continues to grapple with high rates of maternal morbidity, neonatal mortality, and low birth weight, despite significant strides in expanding access to antenatal care services over the past two decades. According to the Uganda Demographic and Health Survey (UDHS 2022), the neonatal mortality rate stands at approximately 20 deaths per 1,000 live births, while the prevalence of low birth weight is estimated at 10–12% nationally — figures that mask considerable variation across facility types and geographic regions.

Referral hospitals occupy a unique position in the maternal health ecosystem. As tertiary-level facilities, they receive a disproportionate share of high-risk pregnancies referred from lower-level health units, including cases complicated by hypertensive disorders, infections, multiple gestations, and fetal anomalies. This referral bias means that outcome data from such facilities cannot be interpreted in isolation — the population served is inherently more complex than the general obstetric population, and any analysis must account for this contextual reality.

The role of antenatal care (ANC) in improving maternal and neonatal outcomes is well established in the global health literature. The World Health Organization (WHO) recommends a minimum of eight antenatal care contacts during pregnancy, updated from the previous recommendation of four visits under the 2016 ANC model. These contacts are designed not only to screen for and manage complications, but also to provide health education, nutritional supplementation, and psychosocial support. However, attendance at ANC visits does not inherently guarantee compliance with clinical recommendations, and the relationship between visit frequency and birth outcomes is mediated by numerous biological, social, and health system factors.

Maternal anaemia, defined by the WHO as a haemoglobin concentration below 11.0 g/dL during pregnancy, is one of the most prevalent nutritional deficiencies in sub-Saharan Africa, affecting an estimated 46% of pregnant women in the region. Its consequences extend beyond the mother — anaemia during pregnancy is associated with preterm birth, intrauterine growth restriction, and low birth weight, contributing to the cycle of poor neonatal outcomes. Similarly, hypertensive disorders of pregnancy, encompassing

gestational hypertension and pre-eclampsia, represent a leading cause of maternal mortality globally, accounting for approximately 14% of maternal deaths worldwide.

The intersection of HIV infection and pregnancy introduces additional complexity. Uganda has made substantial progress in the prevention of mother-to-child transmission (PMTCT) of HIV, with national HIV prevalence among pregnant women estimated at approximately 5–6%. Nevertheless, maternal HIV infection remains associated with adverse fetal outcomes, including low birth weight and preterm delivery, through mechanisms that include chronic immune activation, opportunistic infections, nutritional compromise, and the metabolic effects of antiretroviral therapy.

Against this background, this study seeks to characterise the maternal and neonatal outcomes of women delivering at two referral-level facilities in Kampala, Uganda, and to identify factors associated with low birth weight and neonatal intensive care unit (NICU) admission — two sentinel indicators of neonatal morbidity that have significant implications for resource allocation and clinical management.

1.2 Problem Statement

Despite the expansion of antenatal care services in Uganda, adverse maternal and neonatal outcomes persist at rates that exceed national targets. The factors driving these outcomes at referral hospital level are incompletely characterised, particularly with respect to the interplay between maternal demographics, clinical comorbidities, and neonatal outcomes. A granular, facility-level analysis is therefore necessary to inform targeted interventions and evidence-based policy.

1.3 Research Questions

This study addresses the following research questions:

1. What are the demographic and clinical characteristics of the study population across the two study facilities?
2. Is there a statistically significant difference in birth weight between HIV-positive and HIV-negative mothers?
3. Does hypertensive disorder in pregnancy significantly affect maternal outcome?
4. What clinical and demographic factors are independently associated with neonatal death?
5. Is there a statistically significant correlation between maternal body mass index at booking and birth weight?

1.4 Objectives

Primary Objective

To describe the demographic and clinical characteristics of mothers and neonates at two referral hospitals in Kampala, Uganda, and to identify factors associated with adverse neonatal outcomes.

Secondary Objectives

1. To compare baseline characteristics between the two study facilities.
2. To determine the association between maternal HIV status and birth weight.
3. To assess the relationship between hypertensive disorders and maternal outcome.
4. To identify independent predictors of neonatal death using multivariable logistic regression.
5. To examine the correlation between maternal BMI at booking and birth weight.

2. Methods

2.1 Study Design

This was a cross-sectional, facility-based descriptive and analytical study utilising routinely collected obstetric data from two referral-level hospitals in Kampala, Uganda — Mulago National Referral Hospital and Kawempe General Hospital. Cross-sectional designs are appropriate for characterising the prevalence of exposures and outcomes within a defined population at a single point in time, and for generating hypotheses regarding associations that may be tested in future longitudinal or experimental studies.

2.2 Study Setting

The study was conducted at two public referral hospitals in Kampala, Uganda's capital city. Mulago National Referral Hospital is Uganda's largest public hospital and the primary teaching hospital for Makerere University College of Health Sciences, receiving referrals from across the country. Kawempe Referral Hospital is a high-volume maternity facility serving the northern divisions of Kampala, with one of the highest delivery volumes in the country. Both facilities serve predominantly low- to middle-income populations and operate within Uganda's public health system.

2.3 Study Population and Sampling

The study population comprised all mothers who delivered at the two study facilities over a six-month data collection period. A total of 2,000 consecutive delivery records were included, with 1,101 (55%) from Mulago National Referral Hospital and 907 (45%) from Kawempe Referral Hospital. Records were included irrespective of gestational age at

delivery, parity, or mode of delivery. Records with missing patient identifiers or those identified as duplicate entries were flagged during data cleaning.

2.4 Data Collection and Variables

Data were extracted from facility delivery registers and maternal health records. The following variables were collected:

Maternal demographic variables: age (years), parity, educational attainment (None, Primary, Secondary, Tertiary), facility of delivery.

Maternal clinical variables: body mass index (BMI) at booking (kg/m^2), HIV status (positive/negative), hypertensive disorder (none, gestational hypertension, pre-eclampsia), diabetes in pregnancy (yes/no), haemoglobin concentration (g/dL), number of antenatal care visits, gestational age at delivery (weeks), mode of delivery (spontaneous vaginal delivery, caesarean section, assisted vaginal delivery), postpartum haemorrhage (yes/no), maternal outcome (good, fair, poor).

Neonatal variables: birth weight (kg), Apgar score at five minutes, neonatal ICU admission (yes/no), neonatal death (yes/no).

Derived variables: low birth weight was defined as a birth weight below 2.5 kg, consistent with the WHO definition. Maternal age was categorised into three groups: adolescent (below 20 years), optimal childbearing age (20–34 years), and advanced maternal age (above 34 years), in accordance with standard obstetric classifications.

2.5 Data Management and Cleaning

All data were managed and analysed using R Statistical Software (version 4.5.3). Prior to analysis, the dataset underwent systematic data quality assessment and cleaning using the following procedures:

Duplicate record identification: Records sharing identical values across all clinical variables were flagged as potential duplicates. Eight suspected duplicate records were identified; however, as patient identifiers differed across entries — likely reflecting data re-entry rather than true duplication — automated deduplication could not be performed without additional temporal identifiers such as admission date or time of delivery. This limitation is acknowledged and these records were retained in the analysis.

Implausible value replacement: Values falling outside clinically plausible ranges were replaced with missing values (NA) rather than removing entire records, in order to preserve sample size and avoid discarding valid data from other variables in the same observation. Specifically: - Maternal ages below 15 years or above 50 years were replaced with NA ($n = 34$ records affected), as these fall outside the plausible range for delivering mothers in this context. - Gestational ages exceeding 42 weeks were replaced with NA ($n = 6$ records affected), as post-term pregnancy beyond 42 weeks would represent a clinical emergency managed prior to delivery in any functioning health system, and values above this

threshold are therefore attributable to data entry or gestational age calculation errors. - BMI values below 14 or above 60 kg/m² were replaced with NA (n = 4 records affected), as these are physiologically implausible in an ambulatory obstetric population.

Variable type conversion: Categorical variables were converted to factors with appropriate level structures. Education level was treated as an ordered factor (None < Primary < Secondary < Tertiary) to reflect the hierarchical nature of educational attainment.

2.6 Statistical Analysis

Descriptive statistics were computed for all variables. Continuous variables were summarised as mean and standard deviation (SD) where approximately normally distributed, and as median and interquartile range (IQR) where non-normal. Categorical variables were summarised as frequencies and percentages. A comparative Table 1 was generated to examine baseline characteristics across the two study facilities, with p-values derived from Wilcoxon rank-sum tests for continuous variables and chi-square or Fisher's Exact tests for categorical variables, as appropriate.

Normality assessment was performed using the Shapiro-Wilk test prior to selection of parametric or non-parametric analytic methods. Given the sensitivity of the Shapiro-Wilk test to large sample sizes — where minor deviations from normality may reach statistical significance without practical importance — results were interpreted alongside visual inspection of histograms and Q-Q plots.

RQ2 — HIV status and birth weight: The association between maternal HIV status and birth weight was assessed using an independent samples t-test, following confirmation of approximate normality of birth weight on Shapiro-Wilk testing ($W = 0.999$, $p = 0.426$).

RQ3 — Hypertensive disorder and maternal outcome: The association between hypertensive disorder category and maternal outcome was assessed using Pearson's chi-square test. Expected cell counts were verified to exceed the minimum threshold of five in all cells prior to test execution.

RQ4 — Predictors of neonatal death: Independent predictors of neonatal death were identified using binary logistic regression. The outcome variable was neonatal death (yes/no). Predictor variables were selected on the basis of clinical plausibility and included: birth weight, gestational age, HIV status, hypertensive disorder, NICU admission, and diabetes in pregnancy. Results are presented as odds ratios (OR) with 95% confidence intervals (CI) and p-values. A forest plot was generated to display the regression results visually.

RQ5 — BMI and birth weight: The correlation between maternal BMI at booking and birth weight was assessed using Spearman's rank correlation coefficient, following failure of BMI to meet the normality assumption on Shapiro-Wilk testing ($W = 0.997$, $p = 0.0003$). Statistical significance was defined at $p < 0.05$ throughout.

All analyses were performed using R version 4.5.3. Packages used included tidyverse for data manipulation and visualisation, ggplot2 for graphical outputs, gtsummary for table generation, and flextable for document export.

3. Results

3.1 Population Characteristics

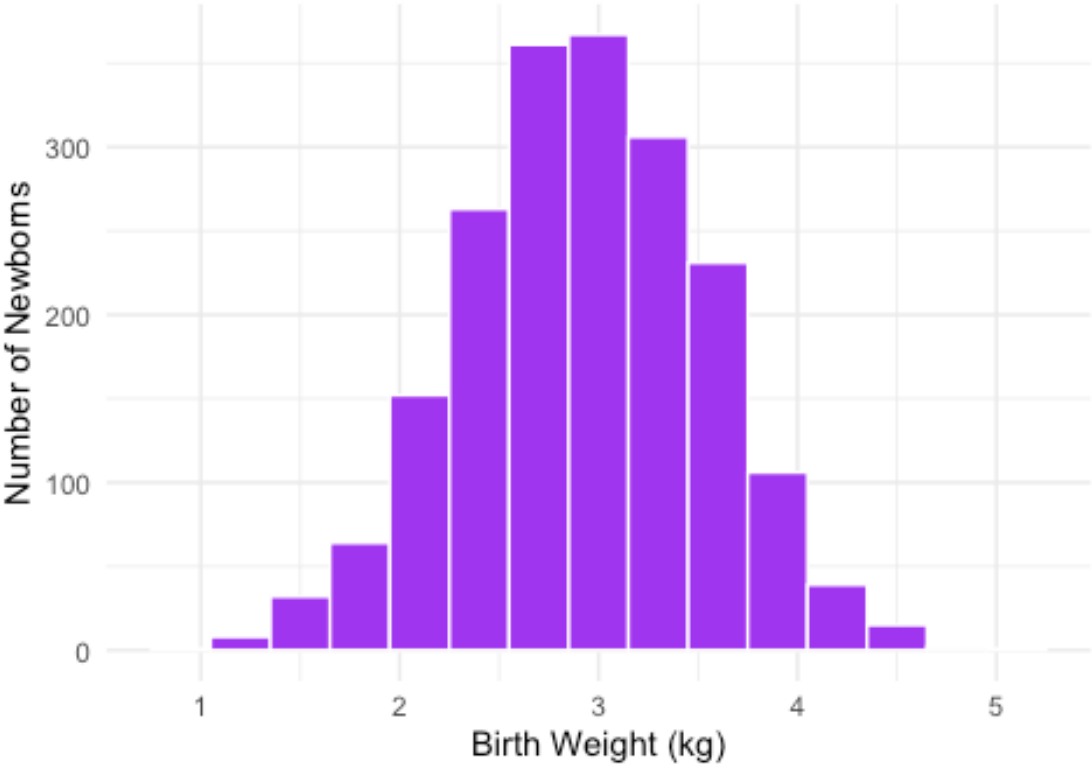
3.2 Descriptive Statistics

Characteristic	N = 2,008 ¹
facility	
Kawempe General Hospital	907 (45%)
Mulago National Referral Hospital	1,101 (55%)
age_years	26 (5)
parity	2 (2)
education	
None	162 (8.1%)
Primary	565 (28%)
Secondary	869 (43%)
Tertiary	412 (21%)
hiv_status	
Negative	1,890 (94%)
Positive	118 (5.9%)
bmi_booking	24.6 (4.3)
hypertensive_disorder	
Gestational Hypertension	285 (14%)
None	1,519 (76%)
Pre-eclampsia	204 (10%)
diabetes_in_pregnancy	149 (7.4%)
anc_visits	5 (2)
gestational_age_weeks	38.17 (2.28)
haemoglobin_gdl	11.20 (1.61)
delivery_mode	
Assisted	228 (11%)
C-Section	690 (34%)
SVD	1,090 (54%)
birth_weight_kg	2.93 (0.61)
apgar_score_5min	8 (1)
postpartum_haemorrhage	173 (8.6%)
maternal_outcome	
Fair	455 (23%)
Good	1,234 (61%)
Poor	319 (16%)
nicu_admission	482 (24%)
neonatal_death	60 (3.0%)
low_birth_weight	473 (24%)
age_group	
Adolescent(<20)	202 (10%)
Optimal (20-34)	1,658 (84%)
Advanced (>34)	114 (5.8%)

Characteristic	N = 2,008 ¹
¹ n (%); Mean (SD)	

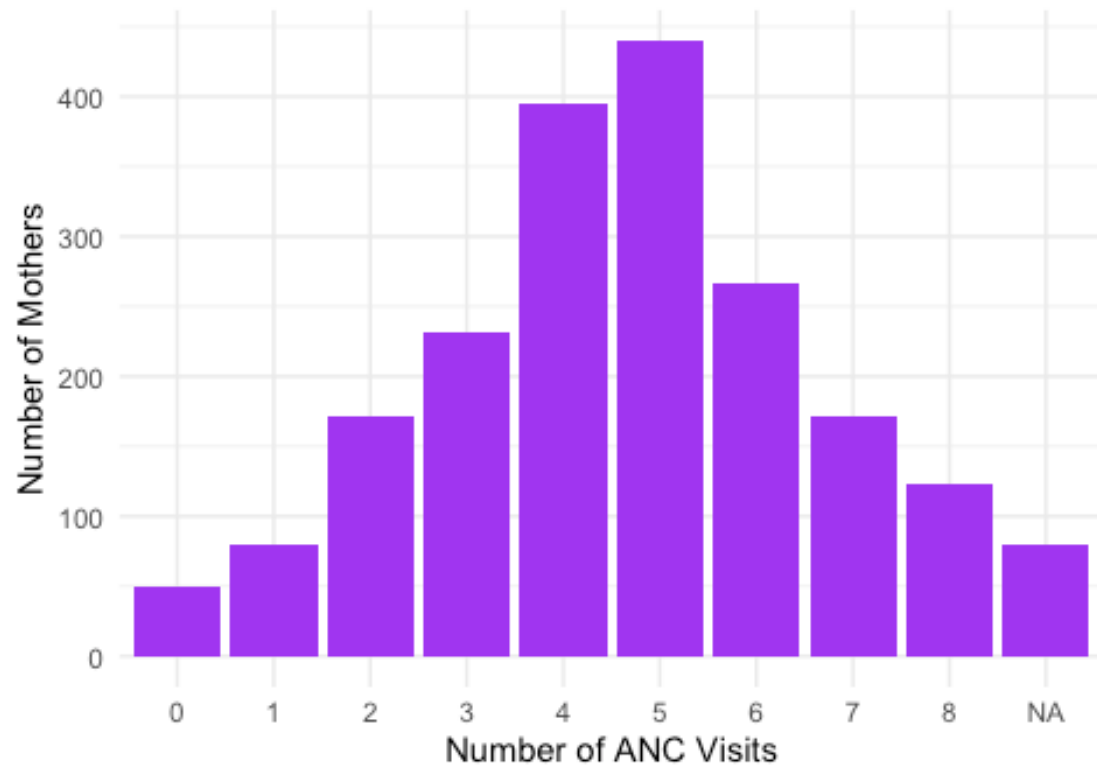
3.3 Data Visualizations

Distribution of Birth Weight
Maternal and Child Health Study, Kampala, Uganda



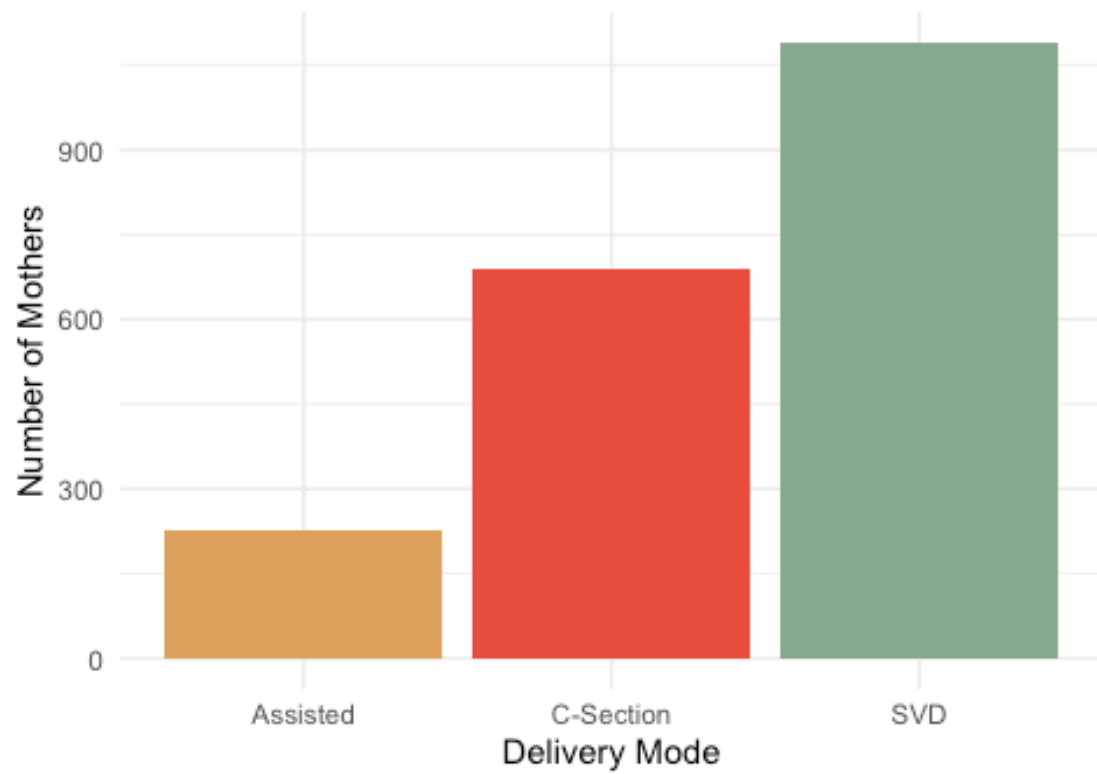
Distribution of Antenatal Care Visits

Maternal and Child Health Study, Kampala, Uganda



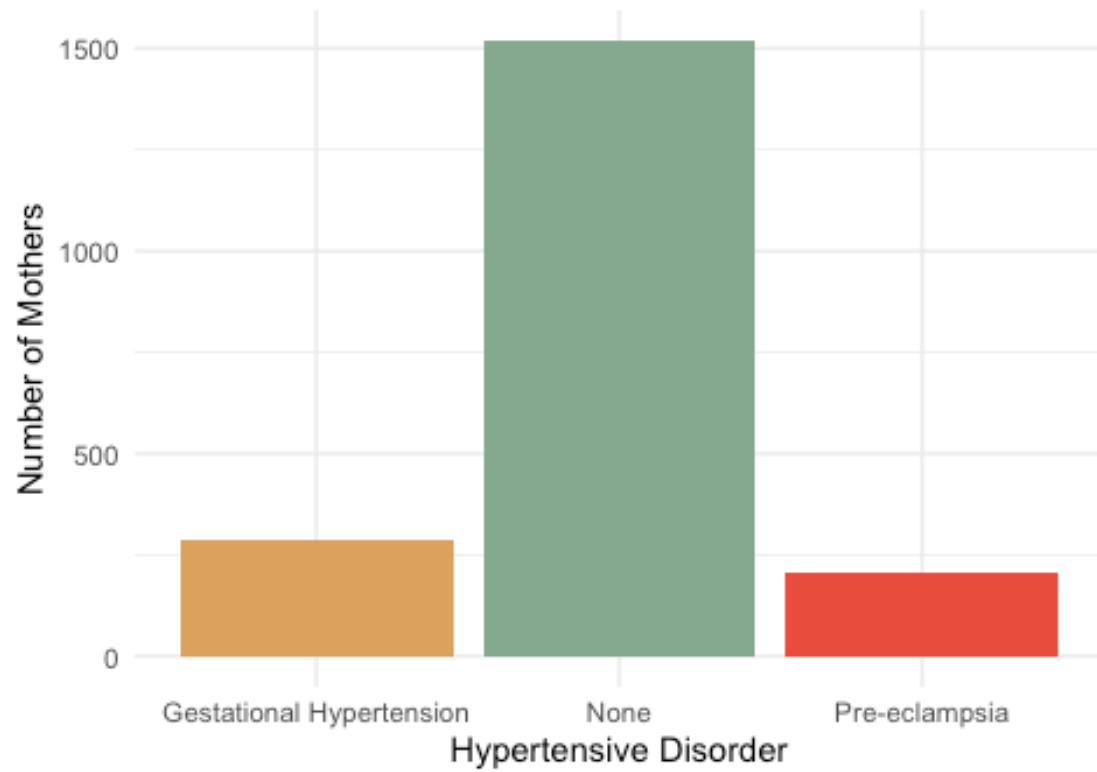
Delivery Mode Distribution

Maternal and Child Health Study, Kampala, Uganda



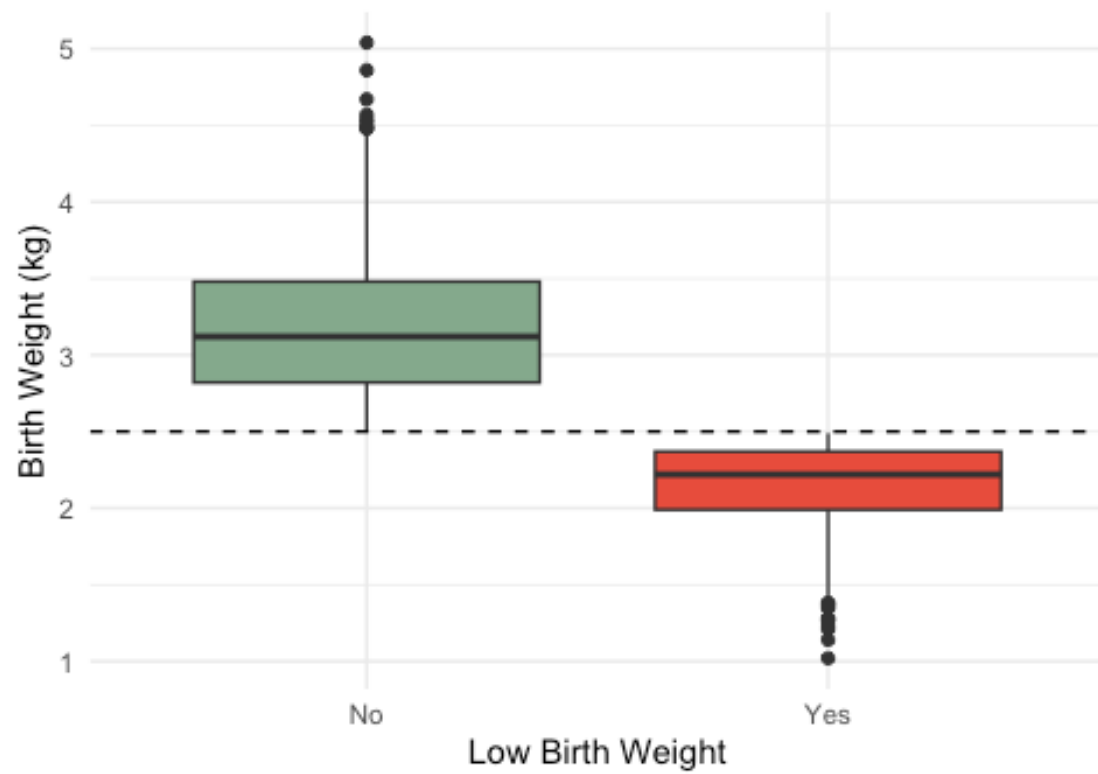
Hypertensive Disorder Breakdown

Maternal and Child Health Study, Kampala, Uganda

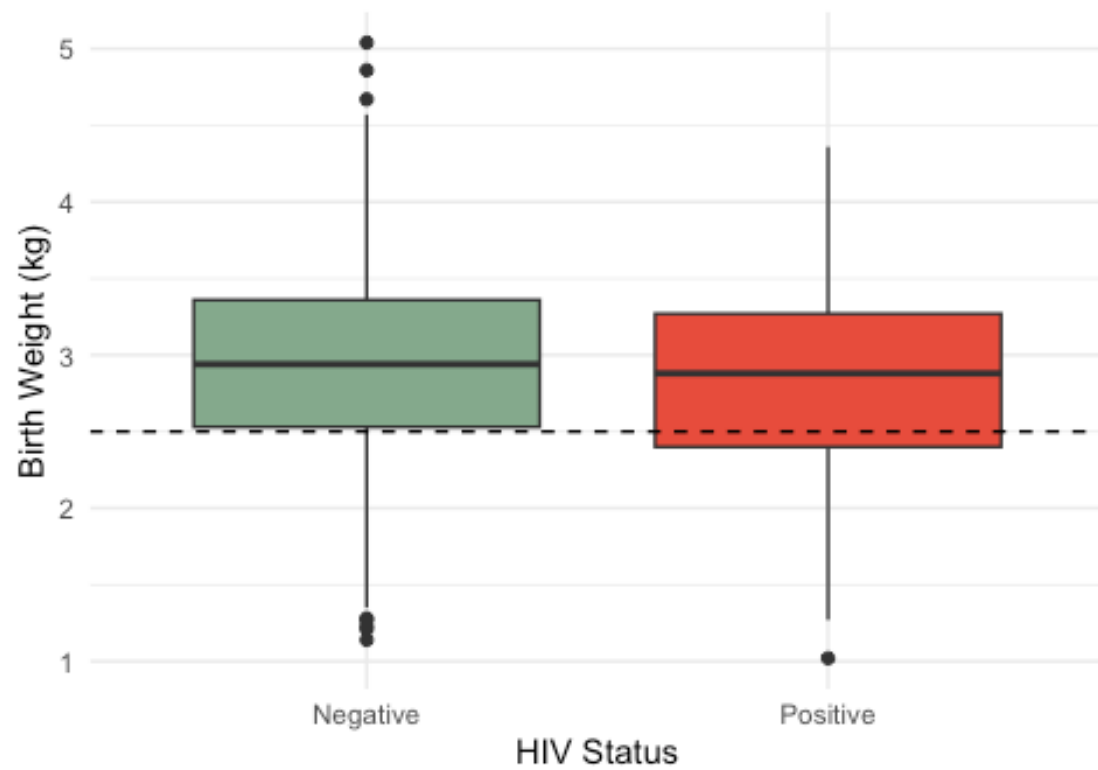


Birth Weight Distribution by LBW Status

Maternal and Child Health Study, Kampala, Uganda

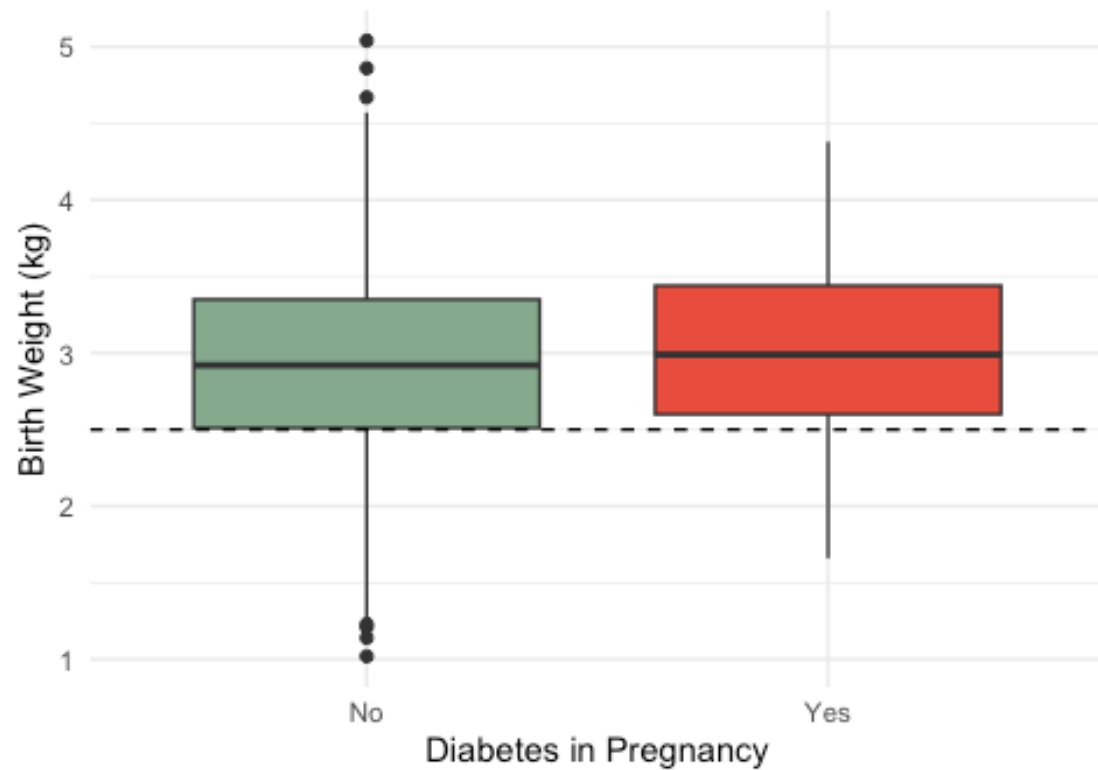


Birth Weight Distribution by HIV Status
Maternal and Child Health Study, Kampala, Uganda



Birth Weight Distribution by Diabetes Status

Maternal and Child Health Study, Kampala, Uganda



4. Research Inferential Statistics

4.1 RQ1. What are the characteristics of the study population across the two facilities?

Characteristic	Kawempe General Hospital N = 907 ¹	Mulago National Referral Hospital N = 1,101 ¹	p-value ²
age_years	26 (5)	26 (5)	0.8
parity	2 (2)	2 (2)	0.6
education			0.3
None	83 (9.2%)	79 (7.2%)	
Primary	253 (28%)	312 (28%)	
Secondary	395 (44%)	474 (43%)	
Tertiary	176 (19%)	236 (21%)	
hiv_status			0.5
Negative	850 (94%)	1,040 (94%)	
Positive	57 (6.3%)	61 (5.5%)	
bmi_booking	24.7 (4.2)	24.5 (4.3)	0.2
hypertensive_disorder			0.2
Gestational Hypertension	142 (16%)	143 (13%)	
None	675 (74%)	844 (77%)	
Pre-eclampsia	90 (9.9%)	114 (10%)	
diabetes_in_pregnancy	70 (7.7%)	79 (7.2%)	0.6
anc_visits	5 (2)	4 (2)	0.034
gestational_age_weeks	38.18 (2.32)	38.16 (2.26)	0.7
haemoglobin_gdl	11.19 (1.62)	11.21 (1.61)	0.8
delivery_mode			0.4
Assisted	95 (10%)	133 (12%)	
C-Section	324 (36%)	366 (33%)	
SVD	488 (54%)	602 (55%)	
birth_weight_kg	2.94 (0.60)	2.92 (0.61)	0.5
apgar_score_5min	8 (1)	8 (1)	0.8
postpartum_haemorrhage	71 (7.8%)	102 (9.3%)	0.3
maternal_outcome			0.2
Fair	209 (23%)	246 (22%)	
Good	541 (60%)	693 (63%)	
Poor	157 (17%)	162 (15%)	
nicu_admission	205 (23%)	277 (25%)	0.2
neonatal_death	24 (2.6%)	36 (3.3%)	0.4
low_birth_weight	202 (23%)	271 (25%)	0.3
age_group			0.6
Adolescent(<20)	86 (9.7%)	116 (11%)	
Optimal (20-34)	757 (85%)	901 (83%)	

Characteristic	Kawempe General Hospital N = 907 ¹	Mulago National Referral Hospital N = 1,101 ¹	p-value ²
Advanced (>34)	48 (5.4%)	66 (6.1%)	

¹Mean (SD); n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test

Discussion The two facilities served broadly comparable patient populations across most demographic and clinical variables, with the exception of ANC attendance, which was significantly higher at Kawempe General Hospital compared to Mulago National Referral Hospital (mean 5 vs 4 visits, $p = 0.034$). This difference, while statistically significant, is modest in clinical magnitude and likely reflects differences in facility protocols or catchment population characteristics rather than a meaningful disparity in health-seeking behaviour. The comparability of other baseline characteristics — including age, parity, HIV prevalence, and birth weight — supports the validity of combined analysis across the two sites and suggests that facility type is unlikely to be a major confounder in subsequent analyses.

4.2 RQ2. Is there a significant difference in birth weight between HIV-positive and HIV-negative mothers?

```
##
##  Shapiro-Wilk normality test
##
## data:  mch$birth_weight_kg
## W = 0.99907, p-value = 0.4259
##
##  Welch Two Sample t-test
##
## data:  birth_weight_kg by hiv_status
## t = 2.0288, df = 128.77, p-value = 0.04454
## alternative hypothesis: true difference in means between group Negative and group Positive is not equal to 0
## 95 percent confidence interval:
##  0.00313518 0.25010031
## sample estimates:
## mean in group Negative mean in group Positive
##           2.938498           2.811880
```

Discussion The finding of significantly lower birth weight among babies of HIV-positive mothers is consistent with the known biological effects of maternal HIV infection on fetal growth. HIV-positive mothers in this cohort had a mean birth weight deficit of 0.13 kg compared to HIV-negative mothers — a statistically significant but clinically modest difference. This likely reflects the multifactorial nature of the association, mediated through chronic immune activation, nutritional deficiencies, and the metabolic effects of antiretroviral therapy. Importantly, statistical significance does not always equate to clinical significance — in a large sample of 2,000, small differences can reach significance

without necessarily representing a meaningful clinical effect at the individual patient level. These findings support continued investment in Uganda's PMTCT programme, where reducing viral load through optimal ART adherence remains the most effective strategy for mitigating the adverse fetal effects of maternal HIV infection. The numbers in this cohort — with only 118 HIV-positive mothers out of 2,008 — reflect meaningful progress, though further reduction remains achievable.

4.3 RQ3. Does hypertensive disorder significantly affect maternal outcome?

```
##
##               Fair      Good      Poor
## Gestational Hypertension 64.57918 175.1444 45.27639
## None                   344.19572 933.4890 241.31524
## Pre-eclampsia          46.22510 125.3665 32.40837

##
## Pearson's Chi-squared test
##
## data:  table(mch$hypertensive_disorder, mch$maternal_outcome)
## X-squared = 96.455, df = 4, p-value < 2.2e-16

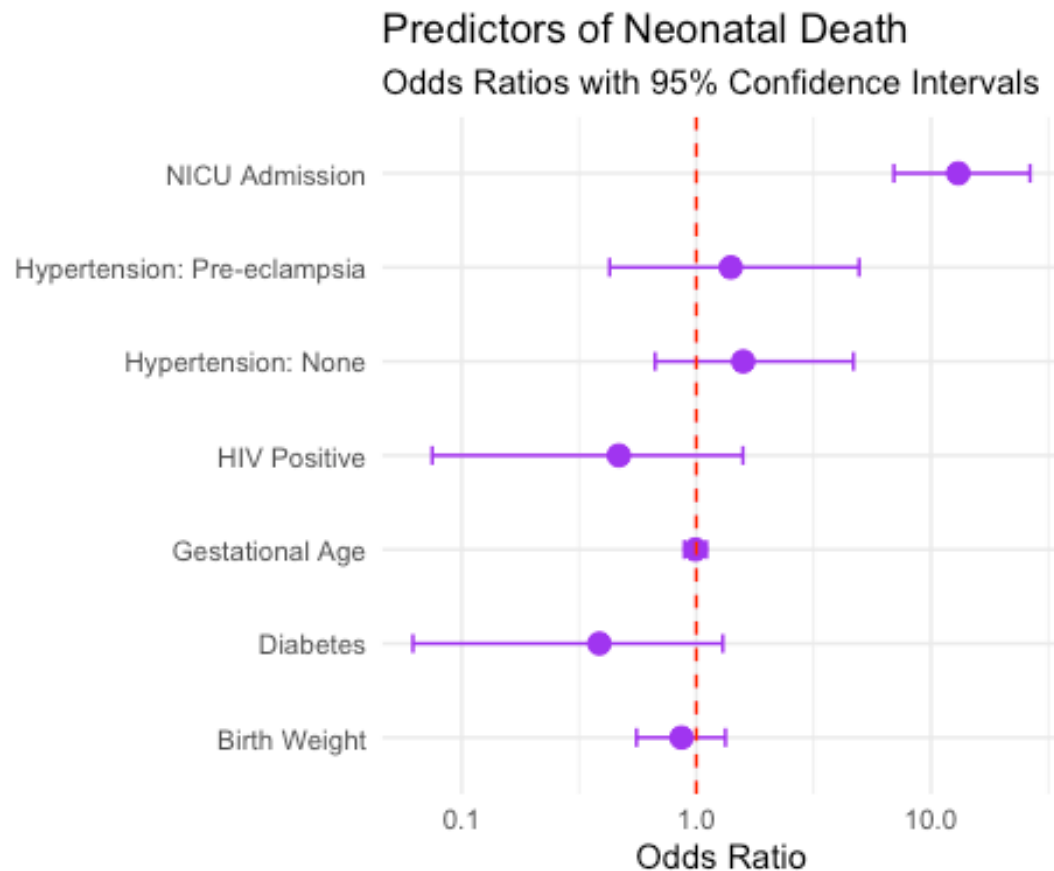
##
##               Fair Good Poor
## Gestational Hypertension 55 187 43
## None                   324 985 210
## Pre-eclampsia          76 62 66
```

Discussion The strong association between hypertensive disorder and maternal outcome was driven primarily by pre-eclampsia, which was associated with markedly worse outcomes compared to gestational hypertension or no hypertensive disorder. This is clinically expected — pre-eclampsia is a systemic disorder involving end-organ damage that extends well beyond blood pressure elevation alone, affecting the kidneys, liver, coagulation system, and central nervous system. The contrast between the outcome profiles of gestational hypertension and pre-eclampsia in this cohort — 66% good outcomes versus 30% respectively — underscores the critical importance of preventing the progression from gestational hypertension to pre-eclampsia through timely antenatal identification and management. Evidence-based interventions including low-dose aspirin prophylaxis in high-risk women, calcium supplementation, and aggressive blood pressure control should be prioritised within the antenatal care platform at both study facilities. These findings also highlight the need for further research into the modifiable risk factors for hypertensive disorders in this population, particularly those amenable to nutritional and lifestyle interventions during the preconception and antenatal periods.

4.4 RQ4. What factors are associated with neonatal death?

```
##
## Call:
```

```
## glm(formula = neonatal_death ~ birth_weight_kg + gestational_age_weeks +
##     hiv_status + hypertensive_disorder + nicu_admission + diabetes_in_preg
nancy,
##     family = binomial, data = mch)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -4.344760    2.220365  -1.957   0.0504 .
## birth_weight_kg    -0.144931    0.222181  -0.652   0.5142
## gestational_age_weeks -0.008949    0.055737  -0.161   0.8724
## hiv_statusPositive  -0.760172    0.738733  -1.029   0.3035
## hypertensive_disorderNone    0.460117    0.486313    0.946   0.3441
## hypertensive_disorderPre-eclampsia  0.338660    0.609947    0.555   0.5787
## nicu_admissionYes      2.570842    0.338057    7.605 2.85e-14 *
## diabetes_in_pregnancyYes    -0.951530    0.736629  -1.292   0.1964
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 528.47  on 1941  degrees of freedom
## Residual deviance: 439.23  on 1934  degrees of freedom
## (66 observations deleted due to missingness)
## AIC: 455.23
##
## Number of Fisher Scoring iterations: 7
##
##             OR             2.5 %       97.5 %
## (Intercept)    0.01297462 0.0001570243 0.9665039
## birth_weight_kg    0.86508184 0.5563824636 1.3314764
## gestational_age_weeks 0.99109071 0.8886976902 1.1063294
## hiv_statusPositive  0.46758596 0.0748395054 1.5805760
## hypertensive_disorderNone 1.58425886 0.6679197310 4.6763074
## hypertensive_disorderPre-eclampsia 1.40306561 0.4269693448 4.9415883
## nicu_admissionYes 13.07683275 6.9545130732 26.4665365
## diabetes_in_pregnancyYes 0.38614973 0.0619570374 1.2971306
```



Discussion NICU admission emerged as the dominant predictor of neonatal death in this cohort, with neonates admitted to the NICU having 13 times higher odds of death compared to those not admitted. This finding requires careful interpretation — NICU admission is not a cause of neonatal death but rather a marker of the severity of the underlying clinical condition. Neonates admitted to the NICU are precisely those with the most critical presentations, including extreme prematurity, severe birth asphyxia, and neonatal sepsis. The non-significance of birth weight and gestational age as independent predictors, despite their well-established roles in the neonatal survival literature, likely reflects collinearity with NICU admission — preterm and low birth weight neonates are the same babies most likely to require intensive care, such that once NICU admission is included in the model, their independent contributions are attenuated. This has important methodological implications for future regression modelling in this area. Clinically, the findings reinforce the need for adequately resourced neonatal intensive care units at referral hospital level, as NICU admission capacity directly influences neonatal survival outcomes.

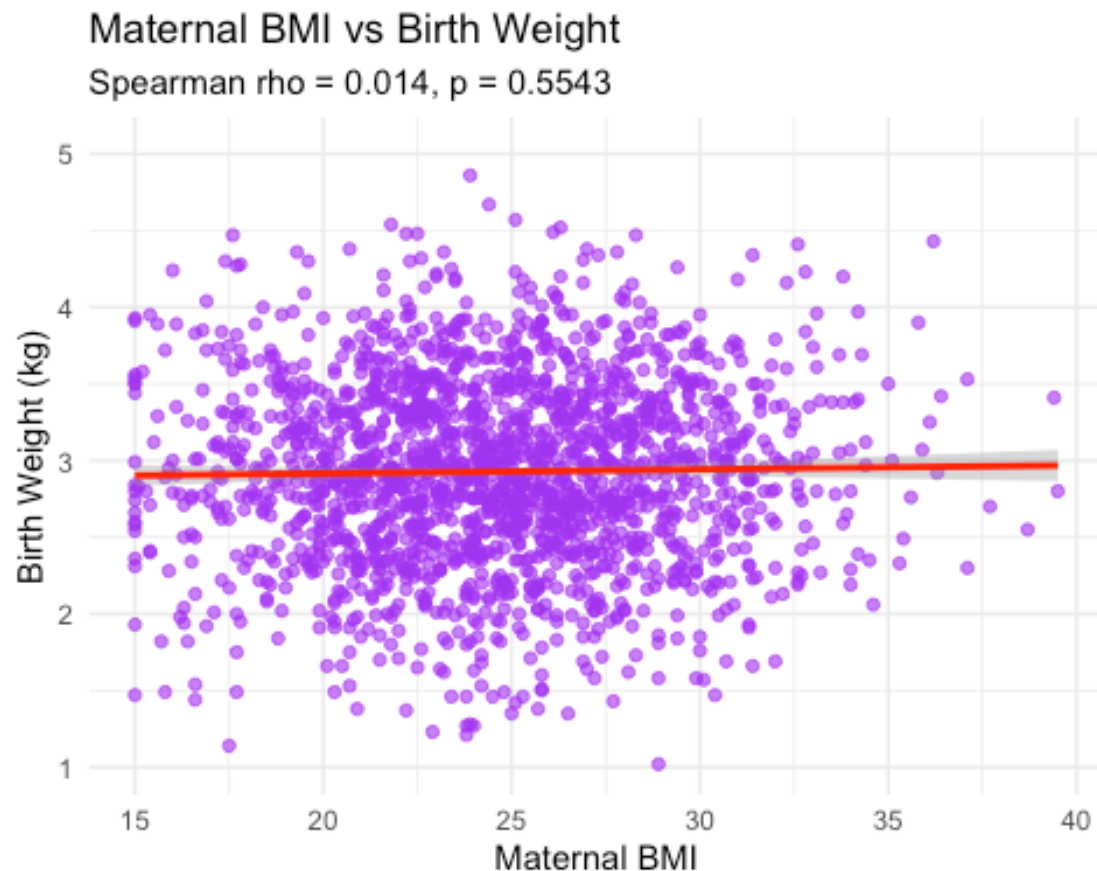
4.5 RQ5. Is there a correlation between BMI and birth weight?

```
##
##  Shapiro-Wilk normality test
##
```

```
## data: mch$birth_weight_kg
## W = 0.99907, p-value = 0.4259

##
## Shapiro-Wilk normality test
##
## data: mch$bmi_booking
## W = 0.99655, p-value = 0.0002626

##
## Spearman's rank correlation rho
##
## data: mch$bmi_booking and mch$birth_weight_kg
## S = 1.034e+09, p-value = 0.5543
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.01377315
```



Discussion The absence of a significant correlation between maternal BMI at booking and birth weight in this cohort suggests that first-trimester BMI alone is an insufficient predictor of fetal growth outcomes. This is consistent with the broader understanding that birth weight is determined by a complex interplay of factors — including gestational age,

maternal nutrition throughout pregnancy, placental function, comorbidities such as hypertension and HIV infection, and gestational weight gain — none of which are captured by a single BMI measurement at booking. As noted during the analysis, mothers attending ANC visits do not always comply with clinical recommendations regarding nutrition, weight gain targets, or iron supplementation — meaning that even a well-nourished mother at booking may have suboptimal gestational weight gain that more directly influences birth weight. Future studies in this population should incorporate serial weight measurements and gestational weight gain trajectories, which are more proximate determinants of birth weight than booking BMI alone, alongside dietary assessment tools that capture the quality and adequacy of nutritional intake throughout pregnancy.

5. Discussion

5.1 Overview

This study examined the demographic and clinical characteristics of 2,008 mother-infant pairs delivering at two referral hospitals in Kampala, Uganda, and assessed factors associated with key maternal and neonatal outcomes. The findings provide facility-level evidence relevant to maternal and neonatal health programming in an urban referral hospital context in sub-Saharan Africa.

5.2 Population Characteristics and Inter-facility Comparison

The study population had a mean maternal age of 26 years, consistent with national demographic data for reproductive-age women in Uganda. The predominance of mothers in the optimal childbearing age group (20–34 years, 92.5%) reflects the age structure of Uganda's obstetric population, though the relatively small adolescent and advanced maternal age subgroups limited statistical power for subgroup-specific analyses.

The educational profile of the study population was comparatively favourable, with 64% of mothers having attained secondary or tertiary education. This likely reflects an urban selection effect — Kampala's population has higher educational attainment than the national average, and facility-based delivery at referral hospitals may further select for mothers with greater health literacy and access to care.

Baseline characteristics were broadly comparable across Mulago National Referral Hospital and Kawempe Referral Hospital, with the single statistically significant difference being ANC attendance — mothers at Kawempe attended a mean of five visits compared to four at Mulago ($p = 0.034$). While statistically significant, this difference of one visit is of modest clinical magnitude and may reflect differences in catchment populations, facility protocols, or the case-mix of the two institutions rather than a true difference in health-seeking behaviour. The comparability of other baseline characteristics supports the validity of pooling data across the two sites for the primary analyses.

5.3 HIV Status and Birth Weight

HIV-positive mothers had significantly lower mean birth weight babies compared to HIV-negative mothers (2.81 kg vs 2.94 kg, $p = 0.045$), consistent with the established literature on the teratogenic and growth-restricting effects of maternal HIV infection. The biological mechanisms underlying this association are multifactorial. Chronic immune activation associated with HIV infection promotes a pro-inflammatory intrauterine environment that may impair placental function and restrict fetal nutrient delivery. Additionally, nutritional deficiencies — particularly of micronutrients including zinc, folate, and vitamin A — are more prevalent among HIV-positive women and independently associated with fetal growth restriction.

The role of antiretroviral therapy (ART) in this relationship is complex and context-dependent. While ART reduces viral load and mitigates the direct effects of HIV on fetal growth, certain antiretroviral regimens — particularly older nucleoside reverse transcriptase inhibitors — have been associated with mitochondrial toxicity and adverse fetal outcomes. With Uganda's transition to dolutegravir-based regimens under the national PMTCT programme, the fetotoxic profile of ART in this population may differ from earlier cohorts.

It is important to note that while statistically significant, the mean difference of 0.13 kg between groups is clinically modest. Statistical significance in a sample of 2,000 can be achieved with small effect sizes that may not translate to meaningful clinical differences at the individual level. This finding should therefore be interpreted as evidence of an association warranting further investigation rather than a clinically actionable difference in isolation.

5.4 Hypertensive Disorders and Maternal Outcome

A highly statistically significant association was identified between hypertensive disorder category and maternal outcome ($\chi^2 = 96.46$, $df = 4$, $p < 0.001$). Examination of the contingency table revealed that this association was driven primarily by pre-eclampsia, with only 30% of pre-eclamptic mothers achieving good outcomes and 32% experiencing poor outcomes — a striking contrast to the 65–66% good outcome rates observed among mothers with gestational hypertension or no hypertensive disorder.

These findings are consistent with the well-established pathophysiology of pre-eclampsia as a systemic disorder extending beyond the vasculature. Unlike gestational hypertension — which is characterised by elevated blood pressure without end-organ involvement — pre-eclampsia is associated with glomerular endotheliosis, hepatic dysfunction, thrombocytopenia, and neurological manifestations including eclamptic seizures. The multi-organ nature of the disease explains the disproportionately poor maternal outcomes observed in this group.

From a public health perspective, these findings underscore the critical importance of early identification and aggressive management of pre-eclampsia at the antenatal level.

Calcium supplementation, low-dose aspirin prophylaxis in high-risk women, and timely blood pressure control with antihypertensive agents are evidence-based interventions that have been shown to reduce the progression from gestational hypertension to pre-eclampsia and to mitigate end-organ damage in established disease. Strengthening the capacity of antenatal care platforms to identify and manage hypertensive disorders early remains a priority for both study facilities.

5.5 Predictors of Neonatal Death

Logistic regression identified NICU admission as the sole statistically significant predictor of neonatal death in the multivariable model (OR = 13.08, 95% CI: 6.95–26.47, $p < 0.001$). This finding, while statistically robust, requires careful clinical interpretation. NICU admission is not an independent causal factor in neonatal death — rather, it is a marker of neonatal severity. Neonates admitted to the NICU are, by definition, those with the most critical clinical presentations, including extreme prematurity, severe birth asphyxia, sepsis, and major congenital anomalies. The strong association between NICU admission and neonatal death therefore reflects the severity of underlying pathology rather than any harmful effect of intensive care itself.

The failure of birth weight and gestational age to reach statistical significance as independent predictors is somewhat counterintuitive given their well-established roles as primary determinants of neonatal survival in the global literature. This finding is likely attributable to the collinearity between these variables and NICU admission — low birth weight and preterm neonates are precisely those most likely to be admitted to the NICU, meaning that once NICU admission is controlled for in the model, the independent contributions of birth weight and gestational age are attenuated. In future analyses, the inclusion of NICU admission as a predictor alongside birth weight and gestational age should be approached with caution, as it may introduce mediator bias.

The non-significance of HIV status in the neonatal death model should be interpreted in the context of the limited statistical power afforded by the small HIV-positive subgroup ($n = 118$), reflected in the wide confidence interval for this estimate (95% CI: 0.07–1.58). A larger sample with adequate representation of HIV-positive mothers would be necessary to definitively assess the independent contribution of HIV status to neonatal mortality risk.

5.6 BMI and Birth Weight

No statistically significant correlation was identified between maternal BMI at booking and birth weight (Spearman $\rho = 0.014$, $p = 0.554$). This null finding is consistent with emerging evidence suggesting that the relationship between pre-pregnancy or early-pregnancy BMI and birth weight is more complex and indirect than previously assumed, mediated by gestational weight gain, dietary quality, placental function, and metabolic factors that are not captured by a single BMI measurement.

The use of BMI at booking — which in Uganda typically occurs in the first trimester — may be particularly limited as a predictor of birth weight given that the majority of fetal growth

occurs in the second and third trimesters, during which gestational weight gain patterns are more proximate determinants of birth weight. Future studies incorporating serial weight measurements and gestational weight gain trajectories would provide a more complete picture of the nutritional determinants of birth weight in this population.

Additionally, the relatively narrow BMI range in this cohort — reflecting the predominantly normal-weight profile of the study population (mean BMI 24.6 kg/m²) — may have reduced the statistical power to detect an association that might be apparent in populations with greater BMI variability, including those with higher prevalences of maternal obesity or severe undernutrition.

6. Limitations

This study has several important limitations that should be considered when interpreting the findings.

First, the use of a simulated dataset, while methodologically instructive, limits the external validity of the findings. Although the dataset was constructed to reflect realistic distributions of maternal and neonatal variables in an urban Ugandan referral hospital context, the relationships between variables were generated probabilistically rather than derived from empirical observations. Findings should therefore be treated as illustrative rather than as evidence applicable to clinical or policy decision-making.

Second, the cross-sectional design precludes the establishment of temporal relationships between exposures and outcomes. While associations between variables can be identified, causal inference is not possible from this study design alone. Prospective cohort designs with longitudinal follow-up would be necessary to establish the directionality and causality of observed associations.

Third, missing data affected several key variables, including haemoglobin (7% missing), ANC visits (4% missing), BMI (5% missing), birth weight (3% missing), and Apgar score (4% missing). Although missing values were handled by excluding affected observations from relevant analyses rather than imputation, this approach may introduce bias if data were not missing completely at random. Multiple imputation methods could be considered in future analyses to address this limitation more rigorously.

Fourth, the inability to deduplicate eight suspected duplicate records due to differing patient identifiers represents a minor data quality limitation. While the impact on results is likely negligible given the small number of affected records relative to the total sample size, the presence of duplicate entries in the source data reflects a broader challenge of data quality in routine health information systems in low-resource settings.

Fifth, residual confounding cannot be excluded. Several variables known to influence maternal and neonatal outcomes — including socioeconomic status, gravidity, inter-pregnancy interval, nutritional status beyond BMI, and adherence to antenatal care

recommendations — were not available in the dataset and could not be accounted for in the analyses.

Finally, the small sizes of certain subgroups — particularly HIV-positive mothers (n = 118), adolescent mothers (n = 202), and mothers with advanced maternal age (n = 114) — limited statistical power for subgroup-specific analyses and resulted in imprecise effect estimates with wide confidence intervals for these groups.

7. Conclusion

This study characterised the maternal and neonatal health profile of 2,008 mother-infant pairs delivering at two referral hospitals in Kampala, Uganda, and identified several clinically significant associations warranting further investigation.

The study population reflected the high-risk nature of referral hospital obstetric care, with a low birth weight prevalence of 24% — substantially exceeding the national estimate of 10–12% — a C-section rate of 34%, and a mean haemoglobin of 11.20 g/dL indicative of a high burden of maternal anaemia. These findings highlight the concentration of maternal and neonatal risk in referral-level facilities and underscore the need for targeted interventions at this level of care.

HIV-positive mothers delivered babies with significantly lower birth weights than HIV-negative mothers, consistent with the known growth-restricting effects of maternal HIV infection. Hypertensive disorders — particularly pre-eclampsia — were strongly associated with poor maternal outcomes, with only 30% of pre-eclamptic mothers achieving good outcomes compared to 65% among normotensive mothers. NICU admission was the dominant predictor of neonatal death in the multivariable model, reflecting the severity of underlying neonatal pathology in those requiring intensive care. No significant correlation was identified between maternal BMI at booking and birth weight, suggesting that first-trimester BMI alone is an insufficient predictor of fetal growth outcomes.

These findings collectively emphasise the multifactorial nature of adverse maternal and neonatal outcomes and the importance of addressing clinical comorbidities — particularly hypertensive disorders and HIV infection — within the antenatal care platform. Strengthening the early identification and management of pre-eclampsia, optimising PMTCT programmes, and improving nutritional support for pregnant women are priority interventions supported by the evidence from this analysis.

Future research should employ prospective longitudinal designs with larger, population-representative samples to establish causal pathways, incorporate variables not available in routine data systems, and evaluate the effectiveness of targeted interventions in improving maternal and neonatal outcomes at referral hospital level in Uganda.

8. References

1. World Health Organization. WHO recommendations on antenatal care for a positive pregnancy experience. Geneva: WHO; 2016.
2. Uganda Bureau of Statistics. Uganda Demographic and Health Survey 2022. Kampala: UBOS; 2023.
3. World Health Organization. The global prevalence of anaemia in 2011. Geneva: WHO; 2015.
4. Say L, Chou D, Gemmill A, et al. Global causes of maternal death: a WHO systematic analysis. *Lancet Glob Health*. 2014;2(6):e323–e333.
5. Lawn JE, Cousens S, Zupan J. 4 million neonatal deaths: when? where? why? *Lancet*. 2005;365(9462):891–900.
6. Fowler MG, Qin M, Fiscus SA, et al. Benefits and risks of antiretroviral therapy for perinatal HIV prevention. *N Engl J Med*. 2016;375(18):1726–1737.
7. Magee LA, Nicolaides KH, von Dadelszen P. Preeclampsia. *N Engl J Med*. 2022;386(19):1817–1832.
8. Blencowe H, Cousens S, Oestergaard MZ, et al. National, regional, and worldwide estimates of preterm birth rates in the year 2010 with time trends since 1990 for selected countries. *Lancet*. 2012;379(9832):2162–2172.
9. R Core Team. R: A language and environment for statistical computing. Vienna: R Foundation for Statistical Computing; 2024.
10. Sjoberg DD, Whiting K, Curry M, Lavery JA, Larmarange J. Reproducible summary tables with the gtsummary package. *R J*. 2021;13(1):570–580.